

Timo Probst

Research Assistant
Chair of Computer Graphics
University of Freiburg, Germany

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EDUCATION

Ph.D. Student <i>Chair of Computer Graphics, Supervisor: Matthias Teschner</i>	University of Freiburg, Germany 2019 – 2021
M.Sc. in Computer Science <i>Graduated with honors.</i> <i>Thesis title: Global Contact Handling and Dry Friction for Particle-based Rigid Bodies.</i>	University of Freiburg, Germany 2019 – 2021
B.Sc. in Computer Science <i>Thesis title: Matrix-free PPE Solver for SPH Fluids</i>	University of Freiburg, Germany 2015 – 2019

EMPLOYMENT

Research Assistant <i>Chair of Computer Graphics</i>	University of Freiburg, Germany 2021 – ...
Student Assistant <i>Chair of Computer Graphics</i>	University of Freiburg, Germany 2019 – 2021
Tutor <i>Python of Biologists, Faculty of Biology</i>	University of Freiburg, Germany 2019

PUBLICATIONS

- [1] Timo Probst and Matthias Teschner. “Unified Pressure, Surface Tension and Friction for SPH Fluids”. In: *ACM Trans. Graph.* 44.1 (Dec. 2024). ISSN: 0730-0301. DOI: 10.1145/3708034.
- [2] Timo Probst and Matthias Teschner. “Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH”. In: *Computer Graphics Forum* 42.1 (2023), pp. 155–179. DOI: 10.1111/cgf.14727.

INVITED TALKS

SIGGRAPH 2025 <i>Unified Pressure, Surface Tension and Friction for SPH Fluids</i>	Vancouver, Canada August 2025
FIFTY2 <i>Simulating Water Droplets with SPH</i>	Freiburg, Germany December 2024
Eurographics 2023 <i>Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH</i>	Saarbrücken, Germany May 2023

REVIEWS

ACM SIGGRAPH	2025, 2026
IEEE Transactions on Visualization and Computer Graphics	2023, 2025, 2026
Eurographics	2026

Symposium on Computer Animation	2026
Journal of Computational Physics	2025, 2026
Nuclear Engineering and Design	2025

GRANTS

Deutschlandstipendium	Germany
<i>Selected based on academic performance and social commitment.</i>	2019 – 2020

SUPERVISION

Master Students

Julian Karrer	University of Freiburg 2026
Joshua Fässler <i>Strong Rigid-Fluid Coupling using a Unified Implicit SPH Solver</i>	University of Freiburg 2024 – 2025
Philipp Sandweg <i>Implicit SPH Simulation of Elastic Solids in a Multi-Material Framework</i>	University of Freiburg 2023 – 2024
Andrés Sandoval <i>Global Matrix-free Pressure Solver for Performant SPH Fluid Simulations</i>	University of Freiburg 2022 – 2023

Undergraduate Students

Marius Hägele <i>Implementation and evaluation of a 2D state equation SPH fluid solver</i>	University of Freiburg 2025 – 2026
Georgij Hergenröder <i>Smoothed Particle Hydrodynamics Fluid Simulation in 2D - Implementation and Evaluation</i>	University of Freiburg 2025 – 2026

SKILLS

Languages: Strong reading, writing and speaking competencies for English, German
Coding: C++, Python, C, C#, OpenCL, SYCL, CUDA
Frameworks: NumPy, Godot, Unreal Engine, Blender